



# FINAL REVIEW

Review the meaning of digital and humanities as they relate to the field of Digital Humanities.

What does the term **Digital** refer to?



## Digital System

A digital system represents information using discrete values.

In digital systems, information is typically encoded in binary form—using combinations of 0s and 1s. However, digital systems can also use other discrete representations, such as multi-level signals or custom symbols, depending on the application.

## Humanities defined in extension

Humanities are often defined in extension by listing a series of disciplines. They encompass a broad spectrum of areas of study, including:

#### ▼ Ancient and Modern Languages

The study of languages spoken in the past and present, focusing on their structure, history, and cultural significance. This discipline explores how language shapes communication, identity, and cultural expression across time and space.

#### ▼ Literature

The analysis, interpretation, and appreciation of written works, including poetry, prose, and drama. Literature studies examine how texts reflect and influence cultural, social, and philosophical contexts.

#### ▼ History

The study of past events, societies, and civilizations through the examination of primary sources, artifacts, and narratives. Historians aim to reconstruct and interpret the complexities of human experiences over time.

#### ▼ Philosophy

The exploration of fundamental questions about existence, knowledge, ethics, and reason. Philosophical inquiry seeks to understand the principles that govern

#### ▼ Religion

The study of belief systems, rituals, and practices that shape spiritual and moral dimensions of human life. This discipline examines the historical, cultural, and philosophical aspects of religions around the world.

#### ▼ Visual and Performing Arts

The study and creation of artistic expressions such as painting, sculpture, music, theater, dance, and film. These arts explore aesthetic, emotional, and social dimensions of human creativity.

#### ▼ Anthropology

The study of human cultures, societies, and biological aspects across time and geography. Anthropology integrates perspectives from archaeology, cultural

studies, and physical anthropology to understand the diversity of human life.

▼ Area Studies

The interdisciplinary examination of specific geographical, cultural, or political regions. Area studies focus on understanding the unique characteristics and dynamics of a particular area, such as Latin America or East Asia.

▼ Communication Studies

The analysis of human communication in various forms, including interpersonal, media, and organizational contexts. This discipline explores how messages are created, transmitted, and interpreted.

▼ Culture Studies

The interdisciplinary study of cultural phenomena, including art, media, ideology, and everyday practices. Cultural studies investigate how culture influences and is influenced by power structures and social identities.

▼ Law

The study of legal systems, principles, and practices that govern societies. This discipline explores the development, application, and impact of laws on individuals and communities.

▼ Linguistics

The scientific study of language, focusing on its structure, function, and evolution. Linguistics examines phonetics, syntax, semantics, and sociolinguistic aspects of human communication.

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**Review the Bateson's definition of information.**



### Information (Bateson)

Information is a difference that makes a difference for an interpreter.

For Bateson, information does not exist in isolation, as in Shannon's bit-strings, but only in relation to a perceiving system that can register a difference and act upon it. A difference in temperature, for instance, remains a mere physical variation unless an organism can detect it and adjust its behavior in response. In the same way, the marks on a page are only contrasts of ink against paper until a reader interprets them as words that carry meaning. Biological evolution offers another illustration: signals in the environment, such as the coloration of an animal, function as information only when they affect survival or reproduction. In all these cases, information is not inherent in the signal itself but emerges through the relation between a difference in the world and a system capable of perceiving and acting upon it. In this sense, Bateson reconciles the history of the concept: information is neither purely digital nor purely analogue, but always **relational, oscillating between discrete signals and continuous interpretation**.

e.g. Recognizing 6 or 9 depends on the perspective of the interpreter (Placed horizontally the 6 and 9 can be read like the cancer zodiac sign)

Bateson's definition of information allows us to draw a direct connection with the process of abstraction, and thereby to define the concept of information processing.

## Review the definition of token and computation



## Token

A token is a concrete instance of a symbol within a system of representation. It materializes an abstract type (for example, the digit “1” or the letter “A”) in a specific medium—such as ink on paper, a sound in speech, or a state in a digital circuit. Tokens are the physical or perceivable carriers that allow abstract symbols to be recognized, manipulated, and transmitted.

### The notion of token is led by the Systems of Representation which emphasis on difference

Systems of representation—composed of elements that guarantee differences that make a difference among each other—are of paramount importance for digital abstraction. Examples include the binary alphabet (0 and 1), written characters in a script, or musical notes in a scale: each element acquires meaning only by being distinct within a structured system.

### Token is to show. The Type-Token distinction (Class and Instance)

This leads to the notion of the token. The term itself comes from Old English *tāc(e)n*, meaning “sign” or “mark,” derived from a Proto-Indo-European root that meant “to show” or “to point out.” Historically, **a token was a visible proof, a small object or symbol that stood for something else**. In philosophy and semiotics, Charles Sanders Peirce later distinguished between types (abstract classes) and tokens (their concrete instantiations). This distinction was then adopted in computing and information theory.

### Tokens ensure the differences between types to be preserved across contexts

In this sense, a token is a concrete instantiation of an element within a system of representation. Whereas the element or type defines the abstract identity (for instance, the digit “1” or the letter “A”), **the token is its actual occurrence in a given medium—a mark on paper, a pulse in an electrical circuit, a sound in speech**. Digital abstraction relies on the reliable reproduction and recognition of tokens, so that the differences between types remain preserved across contexts. Without tokens that can be physically produced, perceived, and distinguished, the system of representation would remain purely abstract and inoperative.

## Tokens in Language Models

In contemporary computer science, the notion of the token plays a central role in language models. Large language models do not operate directly on words or sentences as humans perceive them. Instead, they segment linguistic input into tokens—discrete units that can be efficiently encoded, stored, and manipulated. Depending on the design of the model, a token may correspond to a whole word (as in early systems), to a subword unit or morpheme (as in Byte Pair Encoding), or even to individual characters or bytes. For example, the English word “information” might be split into smaller tokens such as “in,” “forma,” and “tion.” This segmentation ensures that the model can handle both frequent words and rare or novel forms by recombining known tokens.

Thus, in language models, a token is a unit optimized for computation: it must be concrete enough to be encoded as a sequence of numbers, yet abstract enough to represent recurring linguistic patterns. In this way, the original meaning of “token” as a sign that stands for something else remains entirely relevant. Tokens are the carriers that allow abstract language to be transformed into a manipulable digital form.



### Computation

Computation is the systematic manipulation of tokens according to explicit rules. It transforms sequences of tokens into new sequences, step by step, in order to represent, calculate, or simulate processes. Whether performed with stones, fingers, or binary digits, computation always consists in rewriting tokens sequences within a structured system of representation.

## Calculus (pebble) and Digitus (finger)

The very words “calculation” and “digital” carry within them the history of discrete thinking. Calculus comes from the Latin for a small stone, once used as a token to count or keep score. Digital derives from digitus, the finger with which numbers were tallied. Both terms remind us that the digital is, at its core, the art of handling tokens: distinct units (at different level of abstraction, more concrete for stones, a bit less for fingers configurations) that stand in place of quantities or concepts.

Computation emerges as the systematic manipulation of such tokens, a process of transforming symbols step by step.

### **Calculus can be approximated by digital methods**

Their calculus was not digital in form, but it could be approximated by digital methods: rewriting curves as sequences of tokens representing infinitely small intervals, and summing them step by step. Later, numerical analysis would formalize these procedures, showing that the continuous could be simulated by the discrete through careful manipulation of tokens.

The twentieth century crystallized the digital essence of computation. Alan Turing defined the limits of calculability through machines that operate on tokens one step at a time. His formalism captured the very spirit of the digital: **computation as the rewriting of token sequences under finite rules. In this view, every form of knowledge—whether numerical, textual, or visual— could be recoded into tokens and processed algorithmically.**

### **Digital Hardware and “Computation” Software**

Modern computers embody this union. Their hardware is digital: circuits flip between two kinds of tokens, 0 and 1. Every analogue signal must first be sampled, quantized, and encoded as tokens in a binary system. Their software is computation: formal rules that transform sequences of tokens into new sequences, from input to output. Together, they create a universal machine, capable of simulating any process, including those of the analogue world. This is why a digital computer can approximate weather systems, model the human heart, or render a landscape in high fidelity—**it translates continuous processes into discrete sequences of tokens.**

Seen this way, computation is the logic, and the digital is the medium.

Computation describes the rules of transformation; the digital provides the tokens on which those rules operate. Computation finds its most concrete form in the digital, and the digital achieves its universality through computation. The two notions, born from pebbles and fingers, now underpin the very way the modern world processes information.

**Explain an example from the text where methods from the humanities have influenced computer science.**

The study of language provided early models for computation. From the grammatical analyses of Panini in ancient India to the formal grammars of the Renaissance and Enlightenment, the attempt to codify linguistic structures anticipated the rule-based systems of modern programming languages.



The journey of Noam Chomsky is an example of how humanities and computer science intersect. Chomsky established a foundational link between linguistics and computer science by drawing inspiration from Panini, an ancient Indian grammarian who lived around the 5th century BCE. Chomsky himself regarded Panini as his intellectual precursor. Panini's groundbreaking work involved creating a formal system of rules to describe Sanskrit, which stands as the earliest known example of a formal grammatical system (Misra, 2017). This system can be considered a recursive, algorithmic framework for generating and analyzing linguistic structures.

Nearly 2500 years later, Panini's formalism influenced the development of programming languages. Chomsky's way of defining grammar as a set of formal, generative rules—directly inspired by Panini—became a key model for constructing the structure of the ALGOL compiler, one of the pioneering programming languages.

This intertwining of an ancient grammatical proto-system with the foundational structures of modern programming languages exemplifies how ideas in the humanities have shaped key developments in computer science.

#### ▼ Other examples

Aristotle's logic, developed in the context of philosophy, supplied the basic operations of inference that would later be mechanized in circuits and algorithms. Philology and textual criticism also laid important groundwork. Techniques for collating manuscript variants and reconstructing textual traditions prefigured the algorithmic comparison of strings and sequences. Music theory contributed too: the analysis of harmony, counterpoint, and rhythm offered models of structured variation and recursive patterning that resonate with algorithmic thinking. Even rhetoric, concerned with persuasion



and the effective structuring of discourse, can be seen as an ancestor of fields like human-computer interaction design.

## Explain what are regulated representations and the tendencies of their evolution



### Regulated Representation

A regulated representation is a particular case of representation **governed by a set of production and usage rules**. These rules can be intrinsically embedded in the production process of the representation or the result of cultural conventions.

Examples of regulated representation include indexes of names, accounting tables, family trees, flow-chart diagrams, formal processes, and maps of a region.

On the contrary, the production of a sculpture, a painting, or a theatrical play is generally too weakly regulated by conventional rules to be considered an example of a regulated representation.

### Tendencies of evolution: from tools to machines

The general process of this regulating **tendency involves the transformation of conventions into mechanisms**. The regulation usually proceeds in two consecutive steps, mechanizing first the representation's production rules and then its conventional usages. Ultimately, through this process, regulated representations tend to become machine-readable.

### Example of Maps (Production → Usage)

In the case of maps, the mechanization process is begun by a progressive formalization of the recording, gathering, storage, and unification of geographic information. This corresponds to the mechanization of conventional production rules. Paper maps were still produced, sharing similarities with those of the

previous generation, but they were made in a completely different manner. The next stage was the **mechanization of usage conventions**, transforming the regulated representation into machines in which all the possible usages are explicitly treated. The digital maps we use nowadays permit a large set of operations (scaling, rotation, etc.) and offer ways to handle multiple information layers. As machines, they offer many more possibilities than traditional ones. However, these various new modes of usage are explicitly programmed. A paper map can be used freely for purposes other than its original function. In this sense, it is **still a tool**. A digital map can only be accessed through specific input and output commands; it has internalized its own usage rules. **It is therefore a machine.**

## **International Image Interoperability Framework (IIIF) Know what IIIF is and its importance in organizing and sharing cultural heritage**

The International Image Interoperability Framework (IIIF) introduces innovative ways to interact with and organize digital images, redefining traditional static image handling methods. Below are some of its defining features:

Firstly, IIIF replaces the notion of an image as a static file with a client-server relation. This paradigm allows images to be delivered dynamically rather than downloaded in full. For example, an image can be truncated, recolored, rescaled, or otherwise modified on demand. These operations occur on the server side, enabling the requested transformation without altering the original image file or requiring the client to download the entire image beforehand.

Secondly, IIIF introduces the concept of a Manifest, a comprehensive description of the structure and properties of a digital representation of an object. At the Manifest scale, the unit of digitization is defined. Examples include a book containing multiple pages or a box containing a collection of postcards. The Manifest organizes these components into a coherent and navigable format. The list of objects and their relationships within IIIF is captured by the Collection, which serves as a higher-level structure describing the organization of Manifests. Each Manifest, in turn, is composed of Canvases, which function as digital surfaces onto which images, texts, or annotations are overlaid. These layers of

content and annotation (often referred to as Content and Anno) allow for highly flexible and detailed representations of objects.

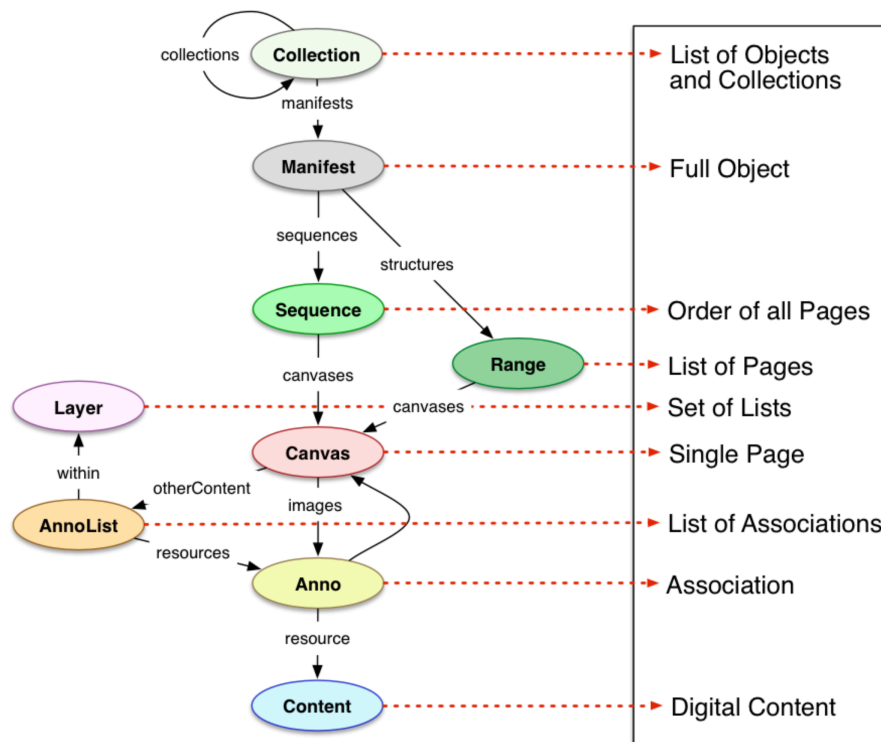


Figure 9.1 IIIF Data structure

Another significant strength of IIIF lies in its open-access and collaborative framework. It is notably easy to ingest other collections into an existing repository. This is accomplished by referencing a "top-node" URL pointing to the desired Collection or Manifest. This seamless integration capability fosters interoperability between institutions and encourages collaborative research efforts.

Example: A museum curating a digital archive of historical manuscripts can integrate relevant materials from another institution's IIIF-compatible collection by simply referencing their Manifest URL. This approach eliminates the need for complex file-sharing processes or physical data transfers while ensuring standardized access to the materials.

# Writing Systems

## Know the difference between a Glyph and a Character



### Character

Components of a script are characters.



### Glyph

The graphical representation of a character is called a **glyph**.

Beware, however, the relationship between glyph and grapheme is not singular, and the way of writing may depend on the context.

### One Character → Multi Glyphs

One character may indeed be represented by different glyphs: depending on the typeface, the letter case, or the handwriting, the same character can take different representations. In some writing systems, different positions within the word or the sentence call for different glyphs – like is the case for some Arabic characters, or for example, with the character sigma (commonly  $\sigma$ , but  $\varsigma$  when at the end of the word).

### One Glyph → Multi Characters

Similarly, a glyph may not correspond to a single character (& is a ligature of “e” and “t” and represents “et” or “and”)

In a way, scripts and writing systems formalizations can be thought of in a similar way as Speech2Text and Text2Speech technologies.

## Greek alphabet vs abjad



It is important to underline the crucial difference between the Greek alphabet and earlier abjad systems such as Phoenician. An abjad records only consonants: to read it, one must already know the language in order to restore the missing vowels. For instance, the sequence of signs "BL" could correspond to "ball," "bowl," or many other possibilities. The act of reading therefore requires a semantic and contextual reconstruction by the reader.

The Greek alphabet introduced a radically different principle: the explicit notation of vowels. This innovation made writing into a code that could be read mechanically, without requiring prior knowledge of the specific words or their semantic context. At first glance, early Greek inscriptions may appear disorderly—lacking punctuation and with variable writing direction—but in essence they mark the birth of an agnostic coding system, in which meaning lies outside the decoding process itself.

With the alphabet, reading became an automatic operation: the combination of signs could be deciphered without interpretative intervention. Where the **abjad was a tool that demanded human completion**, the **Greek alphabet functioned more like a machine that autonomizes reading**. From this point onward, many writing systems pursued this tendency toward greater automatism and rule-governed transcription.

This trajectory stands in contrast with the logographic traditions of Asia, where scripts remained strongly tied to semantics and visual complexity. Yet there are striking exceptions: the Korean writing reform of 1443, when King Sejong introduced Hangul, a fully alphabetic system deliberately designed to replace Chinese characters. This shows that the transition toward self-sufficient, phonetic systems can occur independently, at different places and times.

The history of writing systems is not a straight line but a branching tree, with some branches flourishing into global systems and others disappearing after only a few centuries of use. Among the most influential ancestors is the Phoenician alphabet. Emerging around the 11th century BCE in the Eastern Mediterranean,

Phoenician reduced the complexity of earlier scripts by representing speech with a small set of consonantal signs. This innovation made literacy more accessible, as it required learning only a few dozen symbols rather than hundreds.

From Phoenician, two major lineages emerged:

- The Greek alphabet, which in the 8th century BCE introduced dedicated symbols for vowels. This seemingly small innovation was transformative, as it allowed precise recording of speech and poetry. From Greek, the Latin alphabet developed (via Etruscan), which became the dominant script of Europe and eventually of global communication. Parallel to Latin, the Cyrillic alphabet emerged in the 9th century CE in the Slavic world, also based on Greek forms.
- The Aramaic alphabet, a close relative of Phoenician, spread widely across the Near East. It gave rise to Hebrew, Nabataean (ancestor of Arabic), and influenced writing in Central and South Asia. Some scholars argue that the Brahmic scripts (Sanskrit, Devanagari, Khmer, Thai, Tibetan, etc.) may ultimately derive from Aramaic models, though their exact origin is still debated.

This bifurcation of Phoenician into the Greek–Latin–Cyrillic line and the Aramaic–Brahmic line constitutes one of the great genealogical stories of human culture. Through these branches, a modest set of twenty-two consonantal letters came to underlie the dominant writing systems of Europe, the Middle East, South Asia, and beyond. In many ways, modern alphabets are direct heirs of this ancient simplification, showing how a local innovation could ripple across millennia and continents.

## Understand the concept of operationalization



### Operationalization

An operationalization is the systematic transformation of an abstract concept into a series of concrete operations with measurable outcomes.

Operationalization is the methodological core of computational criticism. It allows abstract notions such as style, genre, or sentiment to be expressed as **quantifiable features** (e.g., sentence length, lexical diversity, syntactic complexity) and tested empirically. The process is neither automatic nor reductive: it requires iterative refinement to align theoretical insight with computational implementation. Through this practice, distant reading becomes a reproducible, data-driven inquiry — one capable of producing falsifiable conjectures in the Popperian sense.

## Know the difference between Panofsky's three level of image interpretation

Close image reading can operate at multiple levels of interpretation. At its most immediate, it involves the perceptual and descriptive recognition of visual forms—the careful inventory of what is seen. At deeper levels, it connects these observations to broader systems of meaning, iconography, and cultural symbolism.

A classical articulation of this multilevel approach is found in Erwin Panofsky's theory of iconological analysis, which distinguishes three successive strata of interpretation:

- Pre-iconographic description — the primary or natural subject matter of an image, involving the direct recognition of forms, gestures, and motifs (e.g., a seated figure holding a book).
- Iconographic analysis — the identification of conventional subject matter, linking visual motifs to textual or cultural sources (e.g., recognizing the figure as Saint Jerome).

- Iconological interpretation — the discovery of intrinsic meaning, or the underlying principles of a culture or epoch that the image embodies (e.g., interpreting the representation as an expression of Renaissance humanism and the contemplative life).

The iconological interpretation is always contextual. The same visual motif—such as

Saint Jerome studying—can acquire entirely different meanings depending on the period, the

patron, and the intended audience. While one painting might emphasize the saint's erudition

and contemplative isolation, another may underline his asceticism, humility, or meditation on

mortality. Behind each image lies an act of commission: a patron who chose to represent a

particular aspect of the saint to convey specific virtues or align with personal, familial, or civic

values. Thus, the image becomes not only a theological symbol but also a social and cultural statement.

In this framework, close image reading functions as the methodological foundation of

all three levels. It begins with attentive looking but ultimately extends toward historical,

philosophical, and cultural understanding. Through this process, the image is treated not

merely as a document but as a coherent field of visual thought, capable of articulating complex ideas through form itself.

In this sense, iconological reading is inseparable from historical empathy. To interpret an image deeply, one must momentarily inhabit the mental and perceptual world of its time— seeing, as it were, through the eyes of its contemporaries.



## Understand the concept of concrete shape and related geometries



### Concrete shape

A concrete shape refers to the actual and precise boundaries of a place at a given moment in time.

This concept captures the geographically and temporally specific outlines that define a spatial entity in its physical form. For instance, the country of France can be understood through a series of **diachronic concrete shapes**, which represent the historical evolution of its territorial boundaries. Each significant political or historical event, such as the signing of a treaty, a territorial annexation, or the loss of land, redefines the concrete shape of France. The Treaty of Verdun in 843 shaped the early partitions of the Carolingian Empire, influencing the territorial outline of what would eventually become France. The 1763 Treaty of Paris ended the Seven Years' War and affected France's overseas territories, altering its global concrete shape.

More generally, concrete shapes can be defined as soon as there is the possibility of a legal dispute (property rights, legal system to be applied, etc.)



### Geometry of a concrete shape

In cartography, the geometry of a concrete shape is a logical combination of polygons that are used to represent a concrete shape in relation to another surface.

Each concrete shape can be approximated by different geometries, which may vary in terms of scale, methods of modelling (e.g., using lines or polygons), and the logical combination of polygons. This is especially relevant when dealing with complex shapes that include holes or disjoint parts.

**Remark:** Each polygon within a geometry is defined by a set of points, known as control points. Some of these control points can be engaged in homologous pairs—linked with other points in other documents. These are then associated with pivot points. In such cases, we can refer to these as homologous geometries.

Geometries could also potentially be defined by other means such as raster contours, splines, or other geometrical definitions.

The concrete shape of a lake's boundary may be irregular, with many inlets and outlets. In a GIS, this shape might be simplified to geometry consisting of a polygon that approximates the lake's outline. This polygon might ignore minor inlets and outlets for the sake of computational efficiency or analytical clarity.

**Consider the state of Texas.** The concrete shape of Texas would include every detail of its border, including minor curves and deviations along rivers or natural formations. However, in a GIS, the geometry representing Texas might simplify these details to a series of straight lines connecting key points (control points) along the border. This simplification helps in data processing and analysis, but may lose some details of the concrete shape.

The case of rivers as boundaries highlights the essential interplay between concrete shapes and geometries in cartography. While concrete shapes provide a real-world basis for geographical boundaries, **the changing nature of features like rivers necessitates continual updates and adjustments to the geometries that represent these shapes in digital mapping systems.** Rivers are naturally dynamic features. Factors such as flooding, erosion, sediment deposition, and human intervention can alter the course of a river over time, changing its concrete shape. The dynamic nature of river-defined boundaries presents a challenge for cartographers and GIS professionals. Maps must be periodically updated to reflect the current course of the river to maintain accuracy. Additionally, the geometries used in digital mapping systems may need to be **simplified or adjusted** to accommodate the changes without overly complicating the map's usability.

**Understand the concept of morphological relation in text systems, image systems, map systems and 3D model systems**



### **Text**

A system of texts based on morphological relationships relates texts for which it is possible to establish that they have a specific morphological motif in common and that one is the source of the other in a sequence of transformations or that they share a common morphological text-source.



### **Image**

A system of images based on morphological relationships relates images for which it is possible to visually establish that they have a specific morphological motif in common and that one is the source of the other in a sequence of transformations or that they share a common morphological image-source.



### **Map**

A system of maps based on morphological relationships relates maps for which it is possible to establish that they have a specific morphological motif in common and that one is the source of the other in a sequence of transformations or that they share a common morphological map-source.



### **3D Model System**

A system of 3D Models based on morphological relationships relates object models for which it is possible to establish that they have a specific morphological motif in common and that one is the source of the other in a sequence of transformations or that they share a common morphological source.

**Understand the difference between concept three concepts :  
Weltanschauung, Zeitgeist, Alltag**



### **Weltanschauung**

Weltanschauung denotes a relatively stable, society-wide schema that structures what counts as real, valuable, and knowable. It comprises cosmologies, axiologies, and legitimation frames.



### **Zeitgeist**

Zeitgeist captures the mid-run orientation of a period: shared sensibilities, narrative frames, and evaluative trends (e.g., romanticism, modernism, austerity, consumer optimism).



### **Alltag**

Alltag denotes the embodied routine: daily practices, objects-in-use, micro-rituals, and spatial-temporal rhythms that materialize the age's orientations.